Universidade de São Paulo

Instituto de Ciências Matemáticas e de Computação (ICMC)

Asymptotics of partition functions in random matrix theory

Synthetic Report for the Masters project, fomented by Fundação de Amparo à Pesquisa do Estado de São Paulo.

Project FAPESP #2024/06997-1

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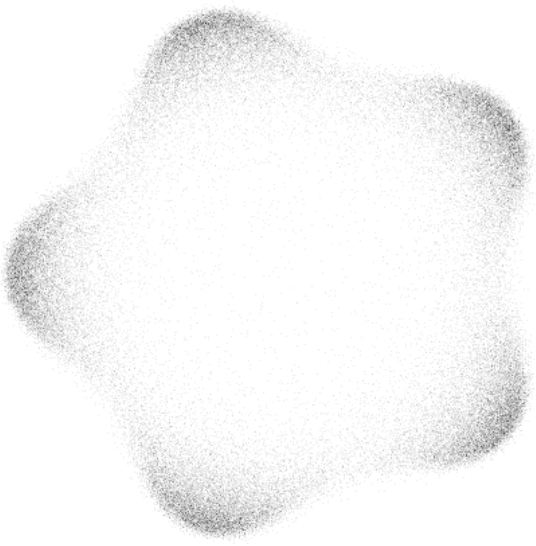

São Carlos, March 18, 2026

Project's General Information

- Project Title:

  Asymptotics of partition functions in random matrix theory

- Supervising Researcher:

  Guilherme L. F. Silva

- Institution:

  Instituto de Ciências Matemáticas e de Computação (ICMC) da Universidade de São Paulo

- Research team:

  Guilherme L. F. Silva

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- Research project number:

  2024/06997-1

- Associated research project number:

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- Project effect dates:

  20/04/2025 a 20/03/2026

- Dates covered by this report:

  01/08/2024 a 31/07/2026

0.1 Activities Overview

The following list summarizes the activities for the months covered by the report.

1. **Completion of the Courses Necessary Credits**: It was completed - in the second semester of 2025 - the necessary courses to fulfill the credits requirements for the Masters course in the Institution. What follows is a transcription of the grades and semesters for the courses taken;

Course	**Code**	**Concept**	**Semester**
Variedades Diferenciáveis	SMA5781-21/3	A	2 - 2024
Equações Diferenciais Ordinárias	SMA5802-17/2	B	2 - 2024
Algebra	SMA5738-17/4	A	1 - 2025
Medida e Integração	SMA5801-13/4	C	1 - 2025
Introdução a Mecânica Celeste	SMA5998-1/1	A	2 - 2025
Preparação Pedagógica	SMA5839-19/8	A	2 - 2025
Treinamento Profissional e Acadêmico	SMA5976-2/1	A	2 - 2025

2. **Publication of papers**: in the time covered by this report two papers were published were the process beneficiary is a co-author. "A Central Limit Theorem for intransitive dice" published in *ALEA, Lat. Am. J. Probab. Math. Stat. (v22-46)*, 2025, with DOI 10.30757. "Simulating Simple Random Walks With a Deck of Cards" published in *Brazilian Journal of Probability and Statistics (25-BJPS635)*, 2025, with DOI 10.1214.

3. **Participation in Events**: Many events have been attended in the period covered by this report. It was presented a poster at **CLAPEM** - Latin American Congress of Probability and Mathematical Statistics held in Montevideo - Uruguay in March, 2026. Money from the research found was used to take part in this event. A poster was also presented at the "XV Workshop de Teses e Dissertações em Matemática" (**WPGMAT**) held in São Carlos - Brazil in October, 2025. He took part in the "Fourier Analysis & Beyond I" (**FAB**) held in IMPA, Rio de Janeiro from June 30th to July 04th, 2025. He also took part in the "35º Colóquio Brasileiro de Matemática" (**CBM**) held IMPA, Rio de Janeiro from July 27th to August 1st, 2025. The student also took part in an extension project called SigmaCamp `https://sigmacamp.org/` - and international advanced scientific camp for young students. He took part in the ETAPA SigmaCAMP and the SigmaFestival where he taught classes on mathematical and physical subjects besides doing administrative tasks along a team of international researchers from all sciences and institutions.

Event	Institution	Date	Modality	Type	Founds
FAB	IMPA	30/06-04/07/2025	Presential	Participant	No
CMB	IMPA	27/07-01/08/25	Presential	Participant	No
WPGMAT	ICMC-USP	01/08/2025	Presential	Poster-Oral	No
CLAPEM	SLAPEM	01-06/03/23	Presential	Poster-Oral	Yes

4. **Continuous study of theory and meaningful literature 0.3**: This M.D. research proposal started in August 2024 and is running for the usual time for the Masters Program, that is, 24 months. The project was originally divided for execution in the following schedule 1. As the natural way to research, we gave continuation to the reading material in the theory and started focusing - in the second semester of 2025 - in the main research article [1] we will work on the remaining months;

Steps	**Quarters**							
	1	2	3	4	5	6	7	8
1. Master Courses	x	x	x	x	x	x		
2. Concepts			x	x	x	x		
3. Main Article				x	x	x	x	
4. Exploration					x	x	x	
5. Masther Thesis					x	x	x	x

Table 1: Activities originally planned. Red color indicates the focus of the quarter that may be divided between activities. Green indicates the planning was accomplished.

Naturally then, until the end of the project the student should be finalizing the study of the main article and writing the final thesis as he explores the possibilities of the research. The writing and organizing of the **Masters Thesis**, what will become the final product of the masters project, was begun. The overview is explicit in 0.3.

0.2 Impact on Student Development

It is of my belief that the masters research project and the course credits associated with the masters program have been crucial in the development of, not only the mathematical tools that make a researcher, but of the mathematical culture. This was specially important coming from physics and computation. Moreover, the work, meetings and discussion completed with the supervisor Guilherme Silva, and the discussions and seminars with the research group were essential in creating a prosperous environment in which the student will develop its own research. Finally, participation in events have, more than the usual familiarization with the possible ramification of his area and particular work, provided a net of peers that will facilitate and encourage future work.

0.3 Research overview and developments

In terms of research we have been trying to discuss the subject of random matrices very broadly. Although the main theoretical work has been on understanding, replicating and studying possible applications of the methods developed in [1], we are trying to approach this as a technique to be used in a larger scenario. The events that the student took part reflect that line of work and are reflecting on the well-roundness of his formation. The work has encompassed creating familiarity with the many direct and indirect applications of Random Matrix Theory (RMT), which encompass extensive fields in mathematics and physics. More directly related to the research, as our research is interested in some asymptotic for a special function related to orthogonal polynomials, we took the time to properly study methods of asymptotics in integrals and the theory of orthogonal polynomials and its connections to the research. In this way, we have been working in the direction of producing the structure and content of the dissertation and expect these themes to compose such text.

To briefly introduce the main subject of our research, we start by the main object. Random Matrices can be though of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such a random matrix. For some Random Matrices, with the help of some mathematical tools such as orthogonal polynomials one can expresses the equilibrium measures for the eigenvalues asymptotically. There are many instances of "models" of random matrices, each one with its symmetries and own probability density. Those are what we call *ensembles*. Borrowing some concepts from physics, an ensemble can be though as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. There is a analogy to be made with ensembles of thermodynamic: When considering sets of matrices with equal probability - in some sense to be more clear soon - we get the Gibbs weight for the probability distribution as we would in the physical *Canonical Ensemble*, thought of as a system of particles with constant temperature. More concretely, in the context of Random Matrix Theory, we have the definition.

Definição 0.3.1 *A random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is a set of matrices $\mathcal{S}$ of fixed size equipped with a probability density function (p.d.f.) $\mathcal{P} : \mathcal{S} \to [0, \infty)$. If we say that a matrix $\hat{X}$ is drawn from the random ensemble $\mathcal{E}$, we are saying that it is a random variable assuming values in $\mathcal{S}$ with p.d.f. $\mathcal{P}$.*

In the research proposed here, we are focusing on the study of *normal random matrices*, which consists of the space of normal matrices, equipped with a proper probability distribution. In the beginning of its book, Feynman states [2] that the key principle of the statistical mechanics is that the probability that a system attains a state with energy E, in equilibrium, is given by a function $\frac{1}{Z}\,\mathrm{e}^{\frac{-E}{kT}}$ where Z is the partition function. In a more general sense there is a relation between the partition function and the free energy of the system, that, in itself, relates to the entropy. Knowing well the partition function is a way to describe with great detail the macro-properties of such a system.

There is a great effort in the community of random matrix theory to study expansions of the partition function of various classes of Random Matrix. Recently, some advance has been made in the works of Sug-Soo Byun et al. [1]. The work they done was on Random Normal Matrix and Planar Ensembles (with Determinantal and Pfaffian structures, respectively), which present partition functions

$$Z_N^{(\beta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^{N} |z_j - z_k|^{\beta} \prod_{j=1}^{N} \mathrm{e}^{-\frac{\beta N}{2} Q(z_j)}\, \mathrm{d}A(z_j) \tag{0.3.1}$$

where N is the size of the matrix, β is relative to the generic entries dimension and $\mathrm{d}A(z) := \mathrm{d}^2 z/\pi$ is the area measure. Here, $Q : \mathbb{C} \to \mathbb{R}$ is called the confining potential and satisfies some suitable conditions: It is smooth, radially symmetric, subharmonic and grows sufficiently fast such that the partition function is finite. In their work, they used the following representation for the partition function

$$\log \mathcal{Z}_N^Q = \log N! + \sum_{j=0}^{N-1} \log h_j, \qquad h_j := \int_{\mathbb{C}} |z|^{2j} \, \mathrm{e}^{-NQ(z)} \, \mathrm{d}A(z), \tag{0.3.2}$$

where h_j are the norming constants for the orthogonal polynomial in respect to the weight function $\mathrm{e}^{-\frac{\beta N}{2} Q(z_j)}$. Then they make an asymptotic analysis using the integral representation for the orthogonality, written in the right. The first step in the asymptotic analysis is to change to radial variables $|z| = r$ and set $V_\tau(r) = q(r) - 2\tau \log(r)$ with $\tau(j) = j/N$. Then, we divide the integral using $\delta_N = \log(N)/\sqrt{N}$ for each h_j, as follows

$$\int_{|r-r_\tau|>\delta} \mathrm{e}^{-NV_\tau(r)} \, 2r\mathrm{d}r + \int_{|r-r_\tau|<\delta} \mathrm{e}^{-NV_\tau(r)} \, 2r\mathrm{d}r.$$

This is typical from the Laplace method - the first term gives the major contribution, the second corresponds to an error term, controlled in the asymptotic analysis. Then, to estimate the summation on $\log(h_j)$, we use the trapezoidal quadrature

$$\int_m^n \rho(x)\mathrm{d}x = \sum_{j=m}^{n} \rho(j) - \frac{\rho(m)+\rho(n))}{2} - \sum_{k=1}^{l} \frac{B_{2k}}{(2k)!}(f^{(2k-1)}(n) - f^{(2k-1)}(m)) + R_l$$

using concepts of orthogonal polynomials and integral asymptotic by Laplace method, they derived large-N asymptotic expansions up to the $O(1)$-terms for $Z_{N,\beta}$ when V is a radially symmetric potential, in the form

$$Z_{N,\beta=2} = -I^V N^2 + \frac{1}{2} N \log N + E^V N + G \log N + F^V + o(1), \quad N \to \infty.$$

In such expansion, for a large class of potentials V the first term I^Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient $1/2$ of the order $N \log N$, as well as the entropy term E^V, were known from recent work of Leblé and Serfaty [3], also for rather large classes of V. The remaining terms in this expansion are new, and they are computed exploring the radially symmetry imposed on the potential. Strikingly, the term G appears to be universal in V, depending solely on the connectivity of the limiting spectrum of eigenvalues. More precisely, they noticed that G depends on whether the limiting spectrum is an annulus or a disc which, surprisingly, seems to not have been observed in the vast previous literature on the matter.

As was said, the study of the partition function is a major way to describe thermodynamic systems. With these new developments it is expected that many systems could be better understood and described by the asymptotic given. We are mainly working now on finalizing the study of the technicalities in the proof for the result above and applying it. We think there is an opportunity here to explore these results in both the light of external fields models and in large deviation theory. Both these application could mean a better understanding of the function behavior on these models, as current techniques are somehow limited.

Bibliography

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[2] FEYNMAN, R. *Statistical Mechanics: A Set Of Lectures*. Avalon Publishing, 1998. (Advanced Books Classics). ISBN 9780813346106. Disponível em: <https://books.google.com.br/books?id=Ou4ltPYiXPgC>.

[3] LEBLÉ, T.; SERFATY, S. Large deviation principle for empirical fields of log and riesz gases. *Invent. Math.*, Springer Science and Business Media LLC, v. 210, n. 3, p. 645–757, dec 2017.